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An Elementary Derivation of the Equivalence of Mass and Energy

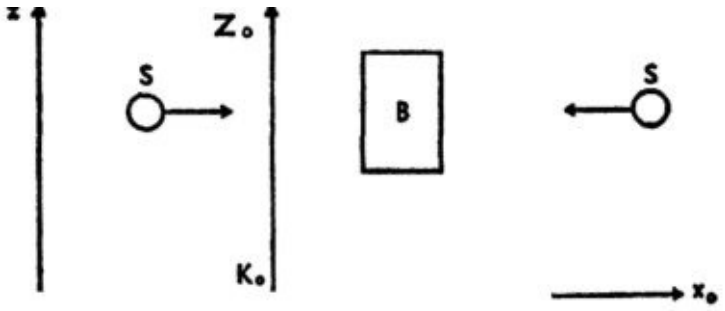
THE FOLLOWING DERIVATION of the law of equivalence, which has not been published before, has two advantages. Although it makes use of the principle of special relativity, it does not presume the formal machinery of the theory but uses only three previously known laws:

(1) The law of the conservation of momentum.

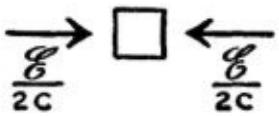
(2) The expression for the pressure of radiation; that is, the momentum of a complex of radiation moving in a fixed direction.

(3) The well known expression for the aberration of light (influence of the motion of the earth on the apparent location of the fixed stars—Bradley).

We now consider the following system. Let the body B rest



freely in space with respect to the system K_0. Two complexes of radiation S, S′ each of energy E/2 move in the positive and negative x_0 direction respectively and are eventually absorbed by B. With this absorption the energy of B increases by E. The body B stays at rest with respect to K_0 by reasons of symmetry.



Now we consider this same process with respect to the system K, which moves with respect to K_0 with the constant velocity v in the negative Z_0 direction. With respect to K the description of the process is as follows:

